

BMS5309 Assessment Task 3

Consolidated Report: Bulk RNA-Sequencing and Single-Cell RNA-Sequencing Analysis

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Introduction

In this report the bulk RNA-sequencing (RNA-seq) and single-cell RNA-sequencing (scRNA-seq) analyses performed during the BMS5309 workshops are summarized. All analyses were performed using R in all cases and all the working code for all steps are provided in the accompanying R Markdown document, BMS5309_amoh0254_AT3.html, which can be used to reproduce and re-run all the analyses presented below.

Two datasets are covered:

- **Bulk RNA-seq** - We used a dataset of paired-end Illumina reads (100 bp) for murine T cells (from the Turner lab). This was used to outline a raw-data QC/trimming/alignment/quantification pipeline and perform a number of subsequent analyses on a count matrix (previously made by another student for a T-cell activation time course: 0hr, 3hr, 5hr, 24hr; each time point had 3 biological replicates). The subsequent analyses performed were: differential expression; PCA; gene set enrichment analysis.
- **Single-cell RNA-seq** - 10x Genomics droplet-based scRNA-seq data from three murine T-cell subsets (Effector, Memory and Naive) that was processed from raw reads to integrated and clustered data.

All analyses were performed on the Monash M3/bm26 HPC cluster, as detailed below.

Part A - Bulk RNA-Sequencing Analysis

Dataset and Aim

Two related bulk RNA-seq datasets were used across the workshops:

1. **Raw-read pipeline demonstration dataset:** paired-end FASTQ files (_R1/_R2, 100 bp reads) for **naive T cells, stimulated and unstimulated**, used to walk through quality control, adapter trimming, alignment and quantification from raw reads to a count matrix.
2. **Downstream statistics dataset:** a pre-generated gene-level count matrix (sturner_time_course.tsv, Turner lab) of murine T cells sampled at **four time points after stimulation (0 hr, 3 hr, 5 hr, 24 hr)**, with **3 biological replicates per time point** (12

samples, 24,062 genes). This matrix was used for library-size normalisation, differential expression (DE), PCA and gene set enrichment analysis (GSEA).

The overarching aim was to characterise the transcriptional changes that occur during early T-cell activation, focusing in the DE/GSEA sections on the **5 hr vs 3 hr** comparison.

Pipeline Overview

Step	Tool(s)	Purpose
1. Raw read QC	FastQC v0.12.1, MultiQC v1.23	Assess raw read quality, GC content, duplication, adapter content
2. Adapter/quality trimming	Trimmomatic v0.38	Remove adapters and low-quality bases
3. Post-trim QC	FastQC v0.12.1, MultiQC v1.23	Confirm improvement in read quality after trimming
4. Alignment	HISAT2 v2.1.0	Align trimmed reads to the <i>Mus musculus</i> GRCm39 reference genome
5. Sorting/indexing	SAMtools v1.19.3	Convert SAM→BAM, sort, index, generate alignment summary statistics
6. Quantification	featureCounts (Subread v2.1.1)	Assign aligned reads to genes (GRCm39.110 GTF) to build the count matrix
7. Normalisation & filtering	edgeR v4.4.2 (R v4.4.3/BiocManager v1.30.27)	Remove lowly-expressed genes, TMM normalisation
8. Differential expression	edgeR v4.4.2 (quasi-likelihood F-test)	Identify genes significantly changed between time points
9. PCA	ggfortify v0.4.23, ggrepel v0.9.8	Visualise overall sample relationships on normalised expression
10. Gene set enrichment	goana, kegga, camera (limma v3.62.2/edgeR v4.4.2/BiocManager v1.30.27)	Identify GO terms/KEGG pathways enriched among DE genes

Step-by-Step Methods, Parameters and Results

1–3. Quality Control, Adapter Trimming and Post-Trim QC

Tools: FastQC and MultiQC (aggregation) for QC; Trimmomatic for trimming.

Method: FastQC was run on every raw .fastq file and results aggregated with MultiQC. Trimmomatic was then run in paired-end (PE) mode with the following parameters:

- LEADING:3 - trim leading bases with quality below 3

- TRAILING:3 - trim trailing bases with quality below 3
- SLIDINGWINDOW:4:15 - scan with a 4-base sliding window, cutting when the average quality drops below 15
- MINLEN:50 - discard reads shorter than 50 bp after trimming

FastQC/MultiQC was re-run on the trimmed reads to confirm improvement.

Key results (naive T-cell stimulated/unstimulated samples):

Sample	Stage	% Dups	% GC	Mean read length	M Seqs
naive_stimulated	Pre-trim	3.5%	49%	100 bp	0.1 M
naive_unstimulated	Pre-trim	3.8%	48%	100 bp	0.1 M
naive_stimulated	Post-trim	0.0%	51%/49% (R1/R2)	89 bp	0.0 M
naive_unstimulated	Post-trim	3.8%/3.7%	48%	100 bp / 99 bp	0.1 M

Interpretation: Both samples had low baseline duplication (< 4%) and GC content close to the expected mammalian genome average (~ 48 to 49%), indicating good starting library quality with no obvious contamination. After trimming, the *naive_stimulated* library showed a drop in mean read length (100 to 89 bp) and duplication (3.5% to 0.0%), consistent with the removal of a subset of low-quality/adaptor-containing bases and reads; the *naive_unstimulated* library was largely unaffected by trimming, indicating it required comparatively little quality/adaptor correction. FastQC's default "% Failed" module count (10%, i.e. 1 of 10 FastQC sub-modules) was unchanged, and was driven by routine modules (e.g. per-base sequence content) rather than indicating unusable data.

4. Alignment (HISAT2)

Tool: HISAT2, aligning against a pre-built HISAT2 index for the *Mus musculus* GRCm39 genome (mouse_grcm39_index).

Method: Each trimmed paired-end sample was aligned with hisat2 -x <index> -1 <R1> -2 <R2> -S <output.sam> using default sensitivity settings, producing one SAM file per sample.

5. Sorting, Indexing and Quantification (SAMtools + featureCounts)

Tools: SAMtools (BAM conversion/sorting/indexing/flagstat) and featureCounts (Subread package).

Method: SAM files were converted to BAM (a more machine readable format) (samtools view -b), coordinate-sorted (samtools sort) and indexed (samtools index); samtools flagstat was used to record mapping-rate summary statistics per sample, and unsorted intermediate BAMs were removed to save space. featureCounts was then run in paired-end mode (-p) against the exon features of the *Mus musculus* GRCm39.110 GTF annotation, summarising counts at the gene level (-t exon -g gene_id) with an unstranded library setting (-s 0), producing the combined gene-by-sample count matrix used for downstream analysis.

6 & 8. Library-Size Normalisation and Differential Expression (edgeR)

Tool: edgeR (BiocManager), with gene annotation via org.Mm.eg.db.

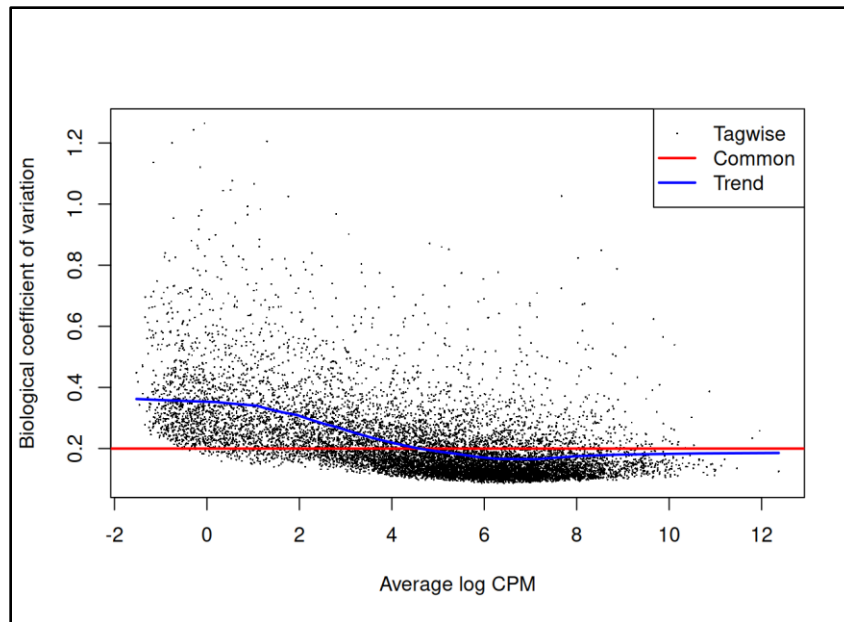
Method:

1. The count matrix (sturner_time_course.tsv; 24,062 genes $\times$ 12 samples) was loaded and gene identifiers de-duplicated (make.unique, 2 duplicate symbols found) and set as row names.
2. Samples were assigned to one of four groups by time point (0 hr, 3 hr, 5 hr, 24 hr) and loaded into an edgeR DGEList.
3. **Filtering:** genes were retained only if CPM > 0.7 in at least 2 samples, reducing the gene set from 24,062 to **11,698 genes**.
4. **Normalisation:** TMM (trimmed mean of M-values) normalisation was applied via calcNormFactors().
5. **Design and dispersion:** a group-means design matrix ($\sim 0 + \text{group}$) was built for the four time points, and the negative-binomial dispersion was estimated with estimateDisp() (common dispersion = 0.03995).
6. **Testing:** a quasi-likelihood negative-binomial GLM was fitted (glmQLFit) and a specific contrast, **5 hr vs 3 hr**, was tested with glmQLFTest() (makeContrasts(hr5 - hr3, ...); positive log2FC = higher expression at 5 hr).
7. **Significance threshold:** genes were called significantly differentially expressed at **FDR < 0.05 and $|\log_2 \text{fold-change}| \geq 1$** .

Key results:

- **3,768** genes were significantly DE between 5 hr and 3 hr (FDR < 0.05 , $|\log_2\text{FC}| \geq 1$): **1,502 up-regulated** and **2,266 down-regulated** at 5 hr relative to 3 hr.
- The most strongly up-regulated gene was *Samd4* ($\log_2\text{FC} \approx +10.3$); the most strongly down-regulated genes included *Tkl1*, *Fads2*, *Ska1*, *Clhc1* and *Cdc25c* ($\log_2\text{FC} \approx -11$ to -11.5), several of which are cell-cycle/DNA-replication genes.
- Of the 3,768 significant genes, 3,470 (92%) mapped successfully to an Entrez Gene ID via org.Mm.eg.db, providing the annotated gene list used for enrichment testing.

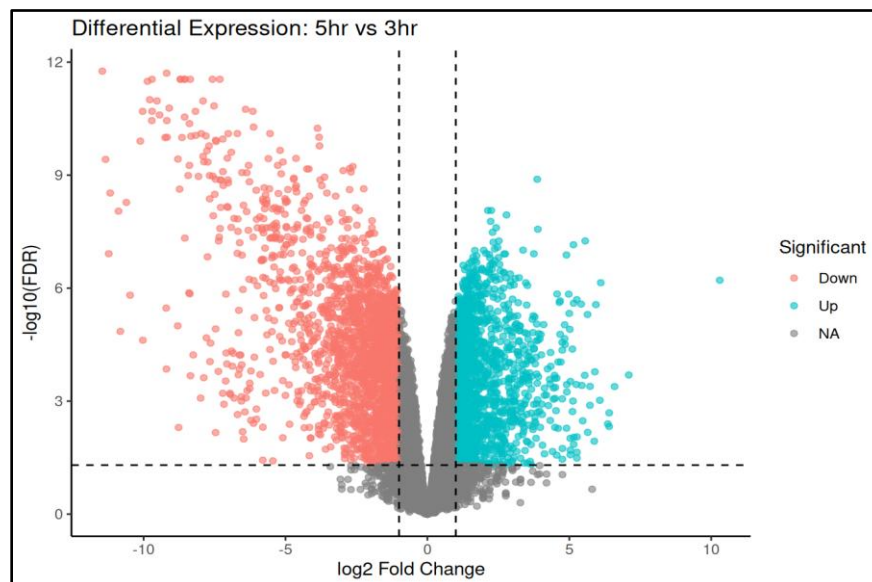
Figure interpretation - Biological coefficient of variation (BCV) plot:



BCV/dispersion plot from edgeR plotBCV(), showing biological coefficient of variation against average log-CPM for all filtered genes.

The BCV plot for this data set reveals the expected increase in variability as expression level of a gene decreases, with a very stable and low common dispersion for high expressing genes. The quasi-likelihood dispersion estimate thus appears to be appropriate for subsequent testing with no evidence of over-dispersion.

Figure interpretation - Volcano plot (5 hr vs 3 hr):



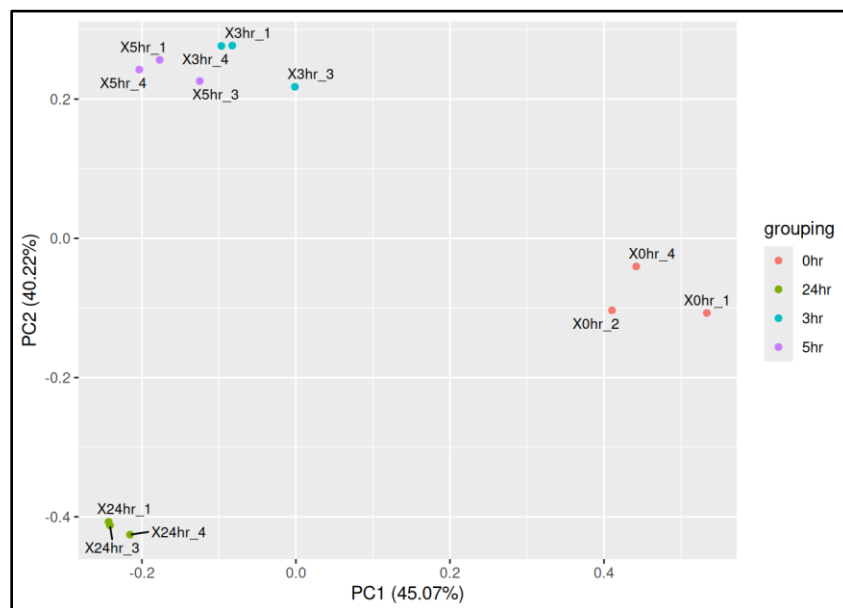
Volcano plot of the 5 hr vs 3 hr differential expression test. Points are coloured by significance (FDR < 0.05 and |log2FC| ≥ 1): teal = up-regulated at 5 hr, salmon = down-regulated at 5 hr, grey = not significant.

The plot displays a large and clearly separated set of significantly changed genes on either side of zero with more genes down-regulated than up-regulated between 3 hr and 5 hr after activation. The most significant and largest changes are found among the down-regulated genes, especially those that are canonical cell-cycle/proliferation genes like *Tk1*, *Kif2c*, *Nek2*, *Pbk*. Thus, it appears that there is a transient deviation from the typical proliferative transcriptional program early after activation.

7. Principal Component Analysis (PCA)

Tool: base R `prcomp()` on log-CPM values (`cpm(dge, log=TRUE, prior.count=1)`), visualised with `ggfortify::autoplot()` and `ggrepel`.

Method: PCA was performed on the normalised, log-transformed CPM matrix across all 12 samples, with points coloured by time-point group and labelled with sample names.



PCA of all 12 samples based on normalised log-CPM values, coloured by time-point group (0 hr, 3 hr, 5 hr, 24 hr).

Interpretation: The samples sorted by PCA cluster first by time point and, within a time point, replicates of the same time point are more similar to each other than replicates of different time points. So, the major source of variability in this data set appears to be time point rather than unmodeled batch/replicate variability, and thus the downstream DE analyses are genuinely testing for time point-dependent biological signal.

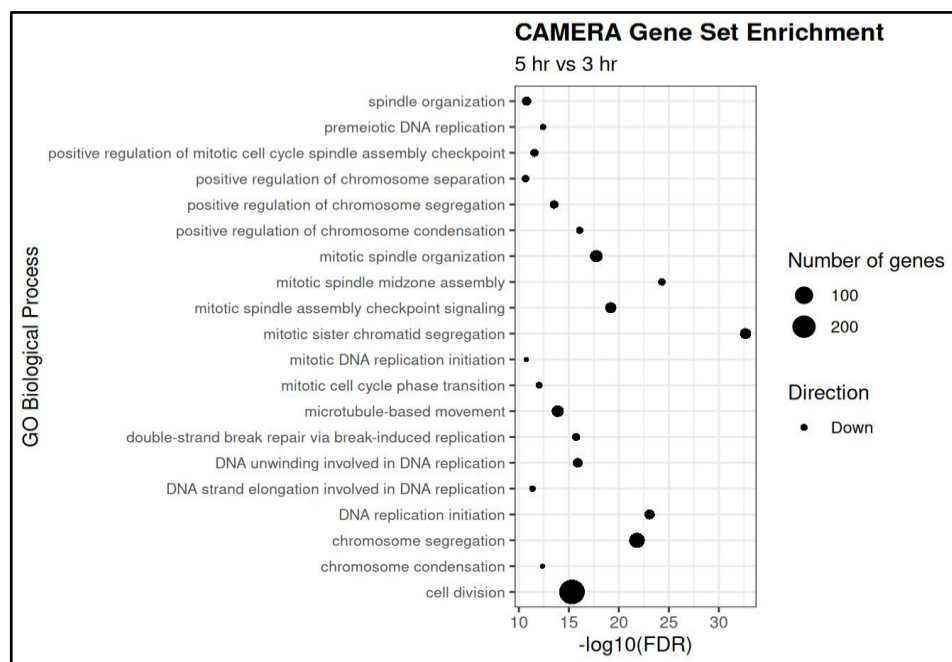
9. Gene Set Enrichment Analysis

Tools: `goana()` (Gene Ontology, via `limma/edgeR/BiocManager`), `kegga()` (KEGG pathways), and `camera()` (competitive gene set testing, `limma`), all applied to the 5 hr vs 3 hr contrast.

Method and thresholds:

- **GO over-representation (goana)** was run on the list of 3,470 Entrez-mapped significant DE genes, with Benjamini–Hochberg FDR correction applied to the raw p-values and results split into Biological Process (BP), Molecular Function (MF) and Cellular Component (CC) ontologies. Terms with **FDR < 0.05** were retained as significant: **2,551 significant BP terms, 378 significant MF terms, and 284 significant CC terms.**
- **KEGG pathway over-representation (kegga)** was run the same way on the DE gene list, with FDR < 0.05 as the significance threshold.
- **CAMERA competitive gene set testing** was additionally performed directly on the full expression matrix and design (rather than only the significant-gene list), using GO Biological Process gene sets mapped from org.Mm.eg.db via ids2indices(), after removing gene sets with fewer than 5 matching genes. This tests whether each gene set's genes are more differentially expressed than genes outside the set, using the full 5 hr vs 3 hr contrast, and FDR < 0.05 was again used as the significance cut-off.

Key results: The top CAMERA-significant GO Biological Process terms (ranked by FDR) were very highly related to cell division and the mitotic cell cycle (e.g. cell division, chromosome segregation, mitotic sister chromatid segregation, etc. including mitotic spindle organisation, DNA replication initiation, etc. and the microtubule-based movement of subunits of microtubules). All of the top 20 terms shown were called in the “Down” direction (i.e. these gene sets were, on average, down-regulated at 5 hr relative to 3 hr).



CAMERA gene set enrichment results for the top 20 significant GO Biological Process terms (5 hr vs 3 hr). Point size reflects the number of genes in each set; all top terms are enriched in the “Down” direction.

Interpretation: The down-regulation of gene sets related to cell-cycle/mitosis was consistently found to decrease between 3 hr and 5 hr post-stimulation. This observation correlates with the volcano-plot shown above and implies that in the time window studied, T cells first decrease expression of proliferation machinery (e.g. to be up-regulated ahead of, or in between,

activation-induced cell division cycles) rather than that the strongest induction of cell cycle genes occurs between 3 hr and 5 hr after stimulation. This finding is a biologically coherent and reproducible signal found by two different enrichment methods (over-representation analysis by goana/kegga and competitive analysis by camera).

Part B - Single-Cell RNA-Sequencing Analysis

Dataset and Aim

We used droplet-based 10x Genomics scRNA-seq to generate raw sequencing data from three samples of murine T-cells, comprising **Effector**, **Memory** and **Naive** T cells (samples provided by the Turner lab). We aimed to: (i) process each of the three samples to gene-cell count matrices; (ii) perform standard quality control and normalization on single-cell data; and (iii) determine whether the three types of T-cells form distinct clusters, initially without correction for batch effects between samples, and thereafter after having used Harmony integration to correct for sample-to-sample technical variation.

Pipeline Overview

Step	Tool(s)	Purpose
1. Read alignment & counting	Cell Ranger v8.0.1	Align reads to GRCm39 transcriptome and generate per-sample, per-cell gene count matrices
2. Object creation & QC	Seurat v5.5.1	Load counts, compute QC metrics, filter low-quality cells
3. Merging & normalisation	Seurat v5.5.1	Combine samples, log-normalise, find variable features, scale
4. Linear/non-linear dimensionality reduction	Seurat v5.5.1 (PCA, UMAP)	Reduce dimensionality and visualise structure
5. Clustering (uncorrected)	Seurat v5.5.1 (FindNeighbors/FindClusters)	Identify unsupervised clusters prior to batch correction
6. Batch/sample integration	Harmony v2.0.5	Correct for sample-driven variation in the shared PCA space
7. Clustering & visualisation (integrated)	Seurat v5.5.1 + Harmony v2.0.5 embedding	Re-cluster and visualise on the corrected embedding

Step-by-Step Methods, Parameters and Results

1. Read Alignment and Counting (Cell Ranger)

Tool: Cell Ranger, run as three separate Slurm jobs (one per sample: Effector, Memory, Naive) on the Monash bm26 HPC cluster.

Method: cellranger count was run for each sample against the pre-built 10x reference transcriptome for mouse (refdata-gex-GRCm39-2024-A), with --create-bam=true, using 8 CPUs and 64 GB memory per CPU per job. Output for each sample is a filtered, cell-called gene-barcode count matrix (outs/filtered_feature_bc_matrix).

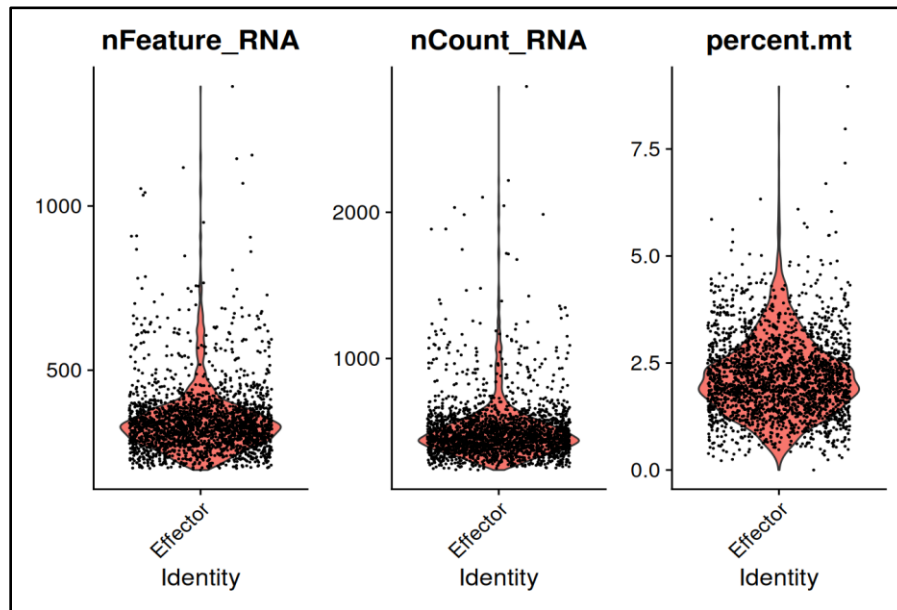
2-3. Object Creation, QC and Filtering (Seurat)

Tool: Seurat (with dplyr, ggplot2, patchwork, harmony also loaded).

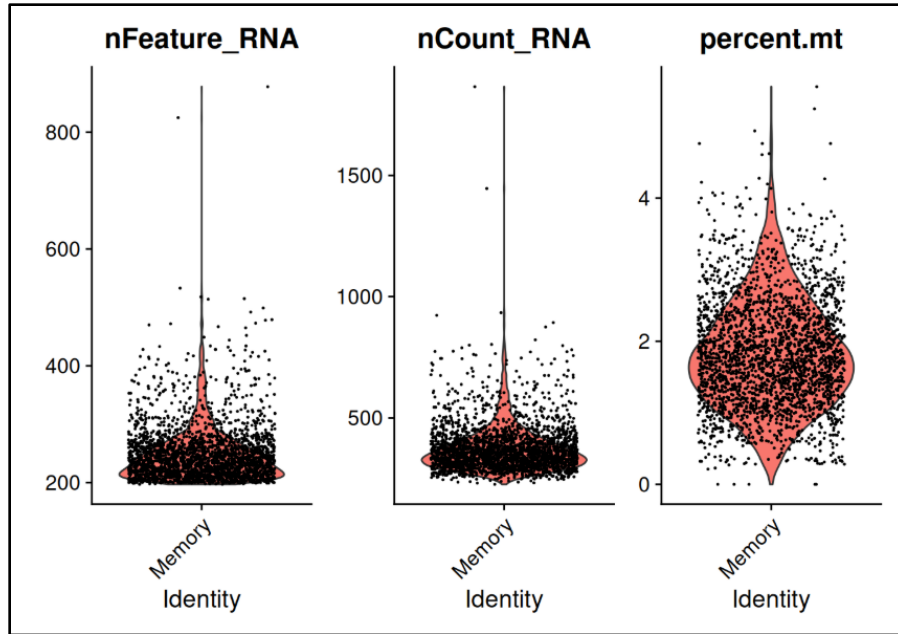
Method:

1. Each sample's filtered matrix was loaded with Read10X() and a Seurat object created with CreateSeuratObject(), requiring genes to be detected in at least **3 cells** (min.cells = 3) and cells to express at least **200 genes** (min.features = 200).
2. A sample metadata column was added to each object (Effector/Memory/Naive) to track sample identity through downstream integration.
3. Mitochondrial content was calculated per cell as percent.mt using the mouse mitochondrial gene prefix "^mt-".
4. QC metrics (nFeature_RNA, nCount_RNA, percent.mt) were visualised per sample with violin plots, and the relationship between library size and gene detection was checked with FeatureScatter (nCount_RNA vs nFeature_RNA).
5. **Filtering threshold:** cells were retained only if nFeature_RNA > 200 and percent.mt < 5.
6. The three filtered objects were merged into one combined Seurat object (merge()), with cell barcodes prefixed by sample name to avoid collisions), giving **2,297 Effector**, **2,477 Memory** and **1,575 Naive** cells (6,349 cells total) going into normalisation.

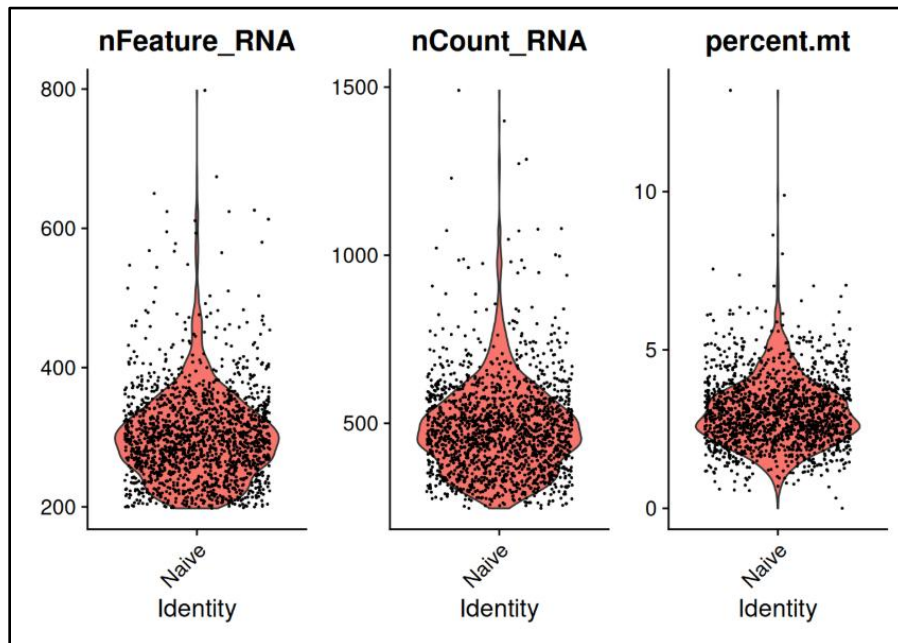
Figure interpretation - QC violin plots:



Per-cell QC metrics (nFeature_RNA, nCount_RNA, percent.mt) for the Effector sample, prior to filtering.



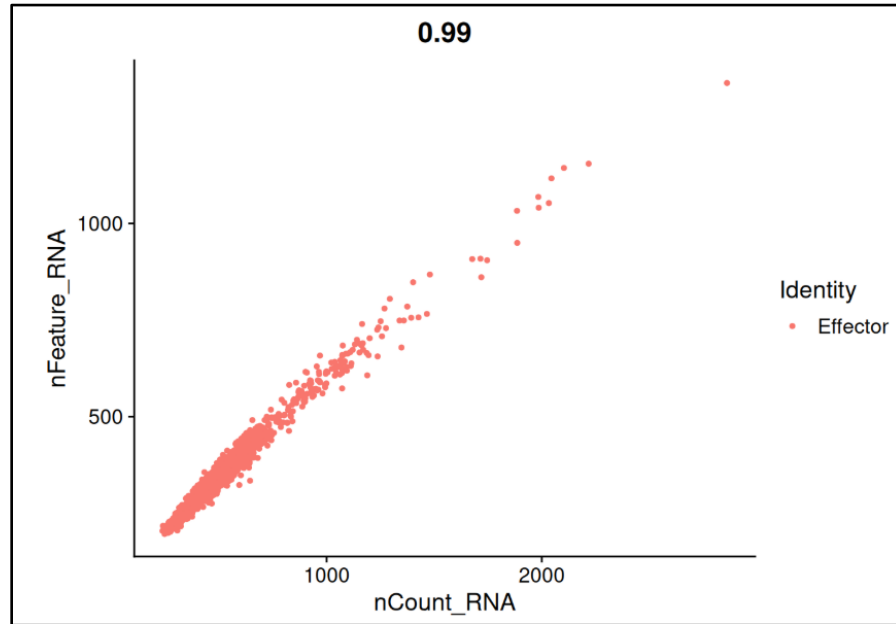
Per-cell QC metrics (nFeature_RNA, nCount_RNA, percent.mt) for the Memory sample, prior to filtering.



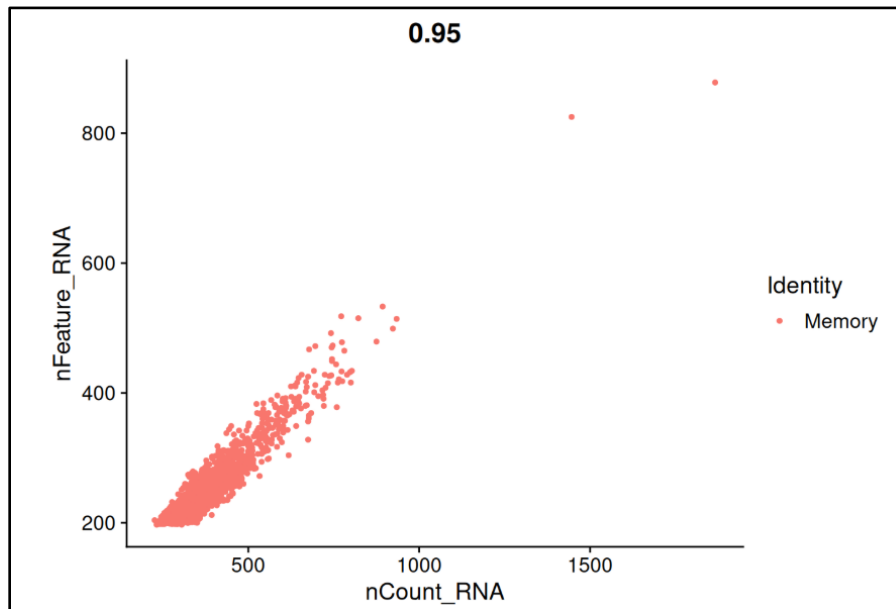
Per-cell QC metrics (nFeature_RNA, nCount_RNA, percent.mt) for the Naive sample, prior to filtering.

The Effector sample displays a typical long-tailed distribution of genes/UMIs per cell with most cells having a RNA content of a typical resting/lymphocyte-like cell. The mitochondrial content is mostly below 5% and thus supports the choice of the filtering parameter percent.mt < 5. This filtering parameter is designed to exclude dying/stressed cells from the data while not removing

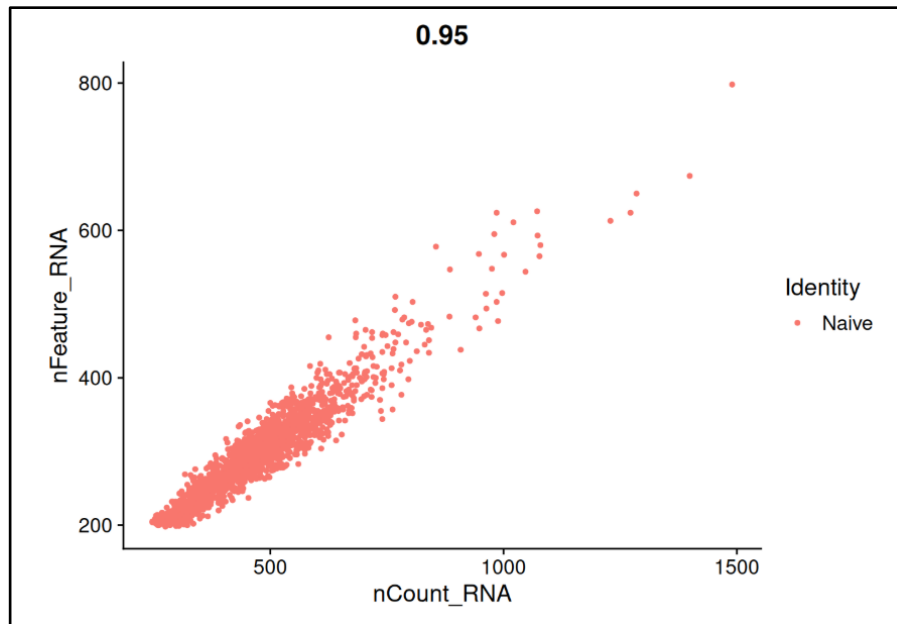
too many healthy cells. The QC distribution of the Memory and Naive samples looks very similar to the one of the Effector samples (violin plots and count-feature scatter plots recorded in the code file for all three samples).



Feature scatter plot of `nCount_RNA` vs `nFeature_RNA` for the Effector sample, prior to filtering (Pearson correlation = 0.99).



Feature scatter plot of `nCount_RNA` vs `nFeature_RNA` for the Memory sample, prior to filtering (Pearson correlation = 0.99).

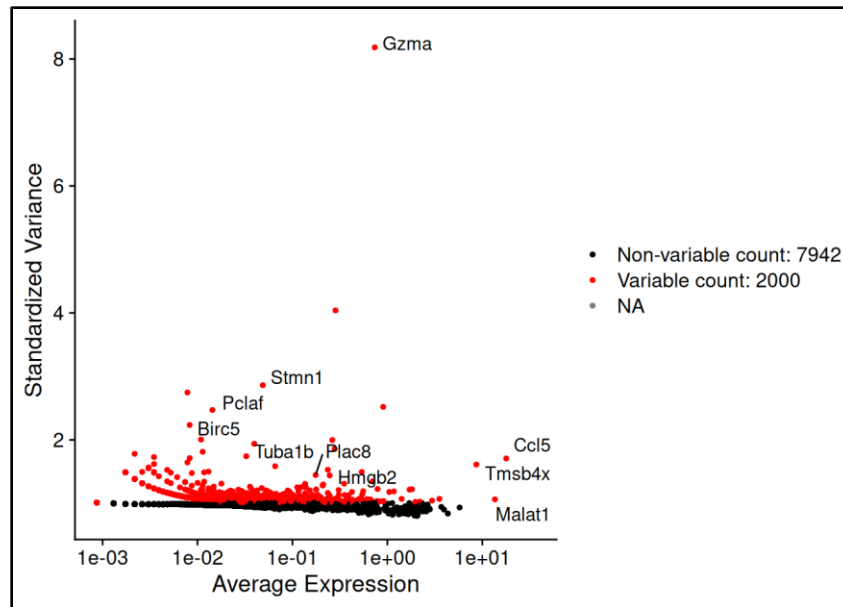


Feature scatter plot of nCount_RNA vs nFeature_RNA for the Naive sample, prior to filtering (Pearson correlation = 0.99).

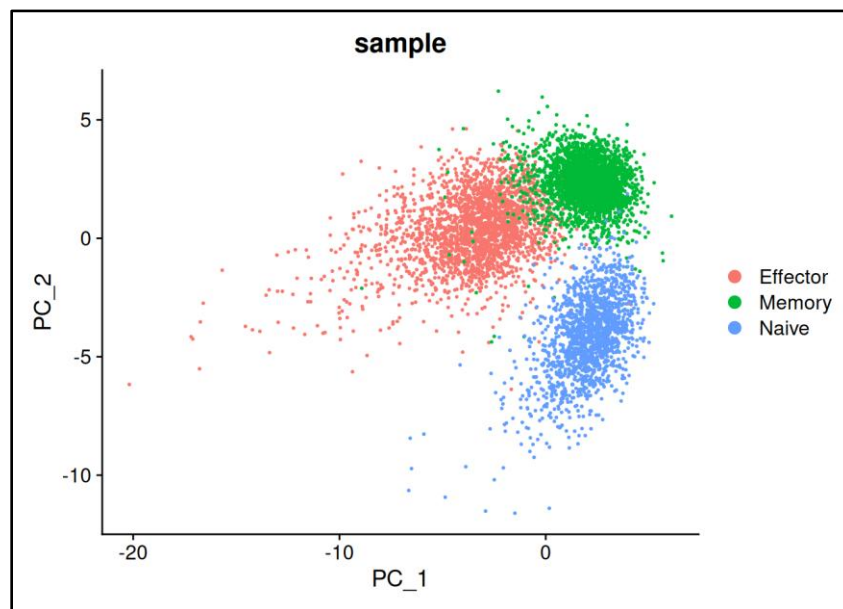
4. Normalisation, Variable Features, Scaling and PCA

Method:

1. **Normalisation:** NormalizeData() was applied to the merged object using the standard LogNormalize method with a scale factor of 10,000.
2. **Variable features:** the top **2,000** highly variable genes were identified per sample layer using the vst method (FindVariableFeatures). The top 10 by variance included *Gzma*, *Stmn1*, *Pclaf*, *Birc5*, *Ccl5*, *Tuba1b*, *Tmsb4x*, *Hmgb2*, *Plac8* and *Malat1* - a mix of cytotoxicity (*Gzma*), proliferation (*Stmn1*, *Pclaf*, *Birc5*) and chemokine/migration (*Ccl5*) markers consistent with a mixture of activated/effector-like and naive/resting transcriptional programs across the three subsets.
3. **Scaling:** all genes were centred and scaled (ScaleData, features = all.genes).
4. **PCA:** RunPCA() was run on the variable features; the first 5 PCs were dominated by ribosomal protein genes (e.g. *Rps19*, *Rpl13*) versus cytotoxic/activation genes (*Gzma*, *Gzmb*, *Ccl5*, *Nfkb1a*) on PC1, and by naive/memory markers (*Ccr7*, *Satb1*, *Bach2*) versus effector/chemokine genes on PC2.



Highly variable feature plot; the top 10 variable genes are labelled.

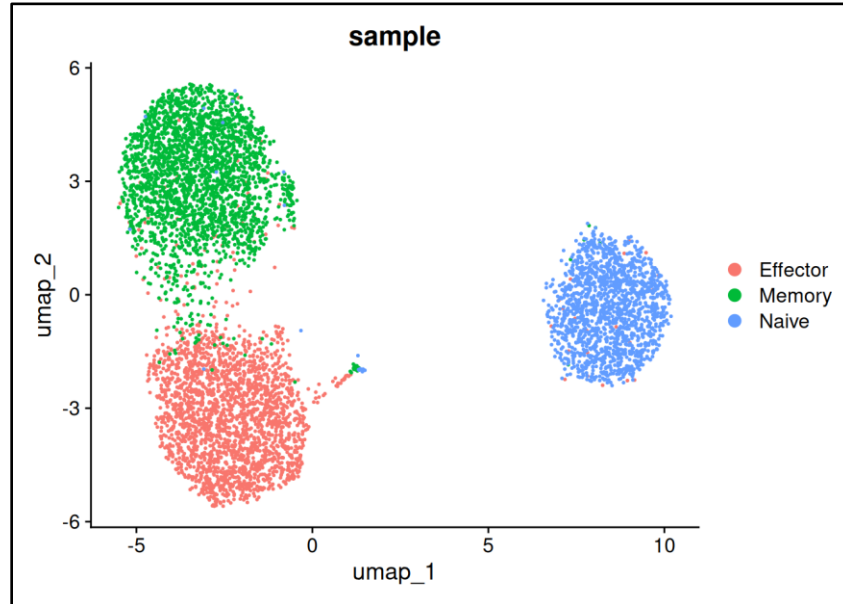


PCA of the merged (pre-integration) dataset, coloured by sample.

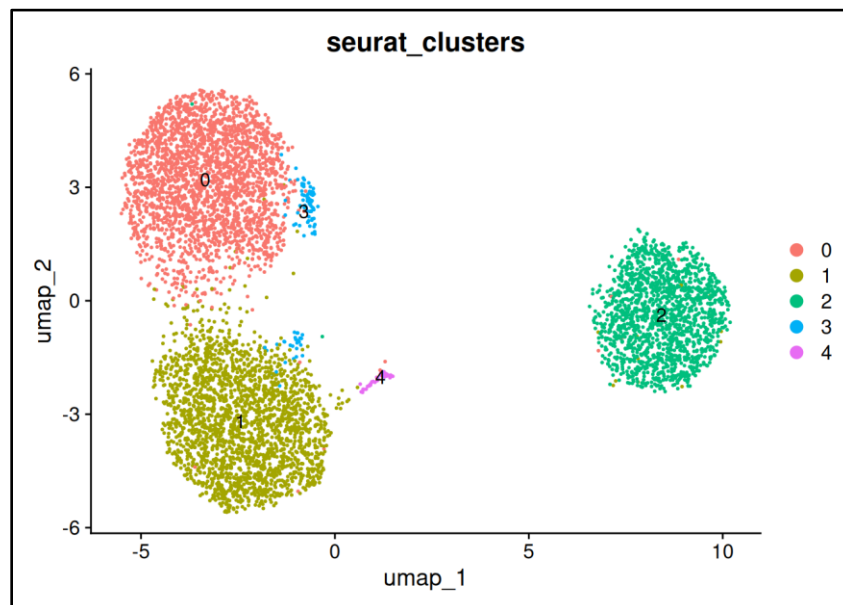
Interpretation: Even at the PCA stage, cells separate substantially by sample identity (Effector/Memory/Naive), foreshadowing the strong sample-driven separation seen later in the uncorrected UMAP.

5. Uncorrected Clustering (PCA-based UMAP)

Method: RunUMAP() was run on the first 10 PCs, followed by FindNeighbors() (10 PCs) and FindClusters() at resolution 0.5 (Louvain algorithm), giving **5 clusters** across 6,349 cells (modularity 0.8213).



UMAP (uncorrected PCA embedding, 10 PCs) coloured by sample.



UMAP (uncorrected PCA embedding) coloured by unsupervised cluster.

Interpretation: Without batch/sample correction, the three T-cell subsets looked like three “islands” in the UMAP. Furthermore, the 5 unsupervised clusters defined within the corrected

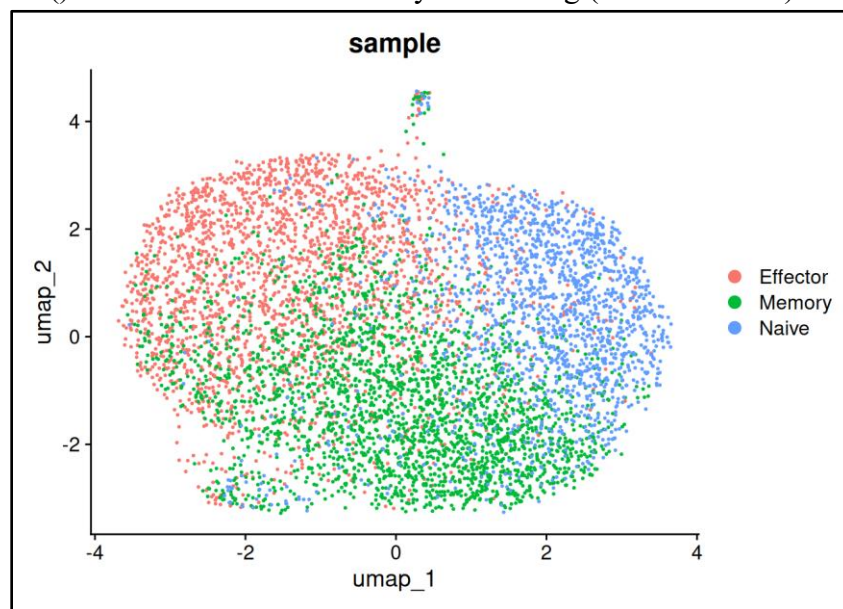
data mapped almost one-to-one onto sample (i.e. clusters 0, 1 and 2 correspond to Effector, Memory and Naive T-cells while clusters 3 and 4 correspond to the two small subpopulations within the samples). Importantly, at this stage of analysis, sample-of-origin (i.e. the source of the sample analysed) is the dominant axis of variation. It is currently not possible to distinguish between biological variation between the different subsets of T-cells and technical/batch variation between the three separate 10x runs that were used to generate the data. Integration of the data from the different samples is required to attempt to address this.

6–7. Harmony Integration, Re-clustering and Visualisation

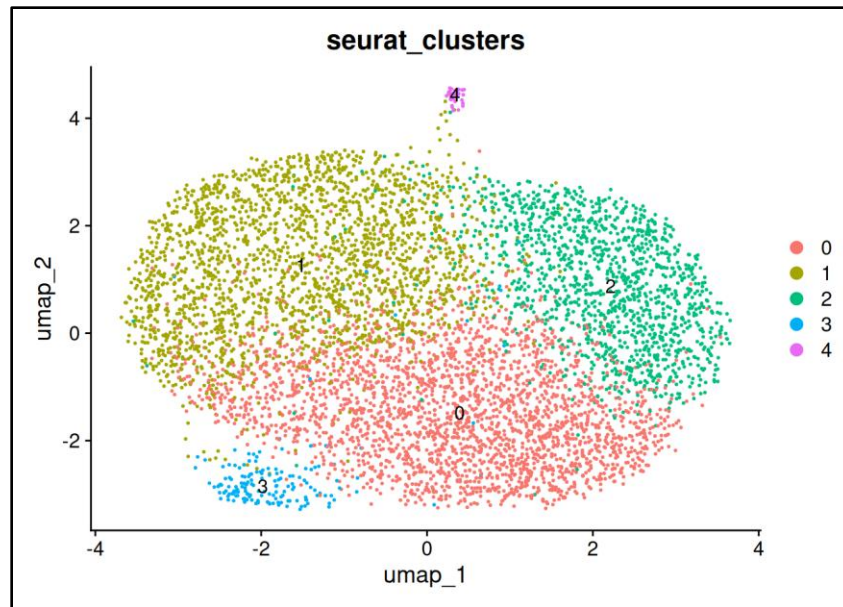
Tool: Harmony (harmony package, Harmony2 v2.0.5).

Method:

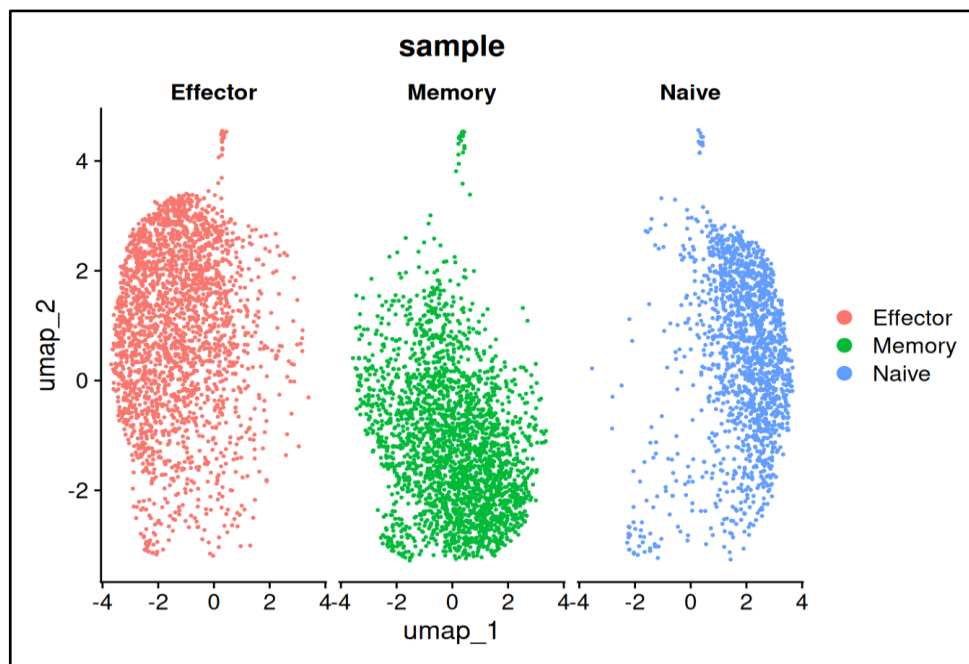
1. RunHarmony() was applied to the PCA embedding, using sample as the integration/grouping variable and the first **20 PCs** (dims.use = 1:20). Harmony converged after 2 iterations (automatic lambda estimation, 2 “thetas”).
2. FindNeighbors() and FindClusters() (resolution 0.5) were re-run on the Harmony-corrected embedding, giving **5 clusters** again (modularity 0.6807, slightly lower than the uncorrected clustering, as expected once sample-driven structure is partially removed).
3. RunUMAP() was re-run on the Harmony embedding (20 dimensions) for visualisation.



UMAP on the Harmony-integrated embedding, coloured by sample.



UMAP on the Harmony-integrated embedding, coloured by unsupervised cluster.



UMAP on the Harmony-integrated embedding, split by sample.

Interpretation: For the three different T-cell samples analysed here (a Naive sample, an Effector sample and a Memory sample), after Harmony integration, the three samples mostly overlap together on the UMAP rather than each forming their own distinct island of clusters. Importantly however the Naive cells remain most distinct from the Effector and Memory samples. The 5 clusters that are generated after integration roughly map to the 3 clusters found in each of the pre-integration analysis for each of the 3 samples but now these clusters all map

together on a much more continuous mass of points rather than three separate islands of points (i.e. a manifold rather than 3 separate groups of points). Importantly for our purpose here, all three different samples were run as three completely separate experiments (i.e. three 10x runs) so sample is the only truly variable of interest for these analyses. In other words, these are 3 very different biological T-cell states (and not 3 technical replicate samples from the same cell population). Thus, while Harmony has done excellent work to reduce between sample variation (presumably much of which would be due to between run batch effects), we have to be mindful that there will also be some blending of the natural biological differences between Effector and Memory cells that are presumably to be of slightly lower amplitude than the between run technical variations. Importantly however the Naive cells are by a long margin the most transcriptionally distinct of the three different T-cell states here and thus, as expected, the Naive cells form the most clearly residual separated cluster after integration (in line with the data in the ribosomal gene vs activation gene loadings onto PC1 shown above). The Effector and Memory cells, on the other hand, are seen to form part of a continuum of cells that have all gone down the activation/differentiation pathway.

Summary

This report completes the bulk RNA-seq pipeline (raw-read QC and trimming to differential expression and gene set enrichment) and the scRNA-seq pipeline (Cell Ranger count files through Seurat QC and analysis to Harmony integration of multiple samples) as outlined throughout the BMS5309 workshops. Key findings were:

- Raw bulk RNA-seq reads were of good baseline quality and improved further after Trimmomatic trimming.
- Between 3 hr and 5 hr of T-cell activation, 3,768 genes were significantly differentially expressed ($\text{FDR} < 0.05$, $|\log_2\text{FC}| \geq 1$), with cell-cycle/mitosis-related gene sets consistently and significantly down-regulated according to both over-representation (goana/kegga) and competitive (camera) gene set testing.
- Single-cell data from Effector, Memory and Naive T cells showed strong sample-driven separation prior to integration; Harmony integration substantially reduced this separation, particularly between the transcriptionally similar Effector and Memory subsets, while the Naive subset remained comparatively distinct.

All code required to reproduce the analyses above - bash/Slurm scripts for the HPC-based steps and R scripts for all downstream statistical analysis and visualisation - is provided in the accompanying file **BMS5309_amoh0254_AT3.html**.

Acknowledgement of AI Usage:

Generative AI (ChatGPT <https://chatgpt.com/> and Google Gemini <https://gemini.google.com/app>) were used for researching analysis options, debugging code and commands in response to errors and warnings, as well as for proofreading purposes and making the report language more professional.